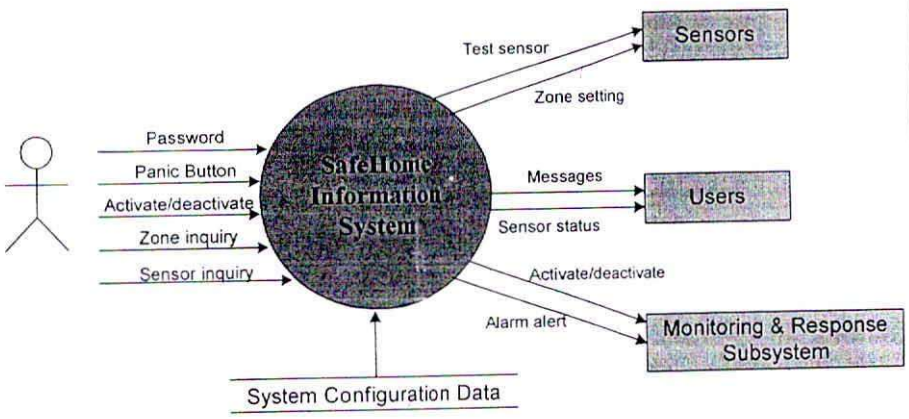
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous Institute Affiliated to University of Mumbai)

ESE Examination

Max. Marks: 100 December 2023 Duration: 3Hrs.
Class: T.E. Semester-V Computer Engineering Course Code: CS301
Name of the Course: Software Engineering

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q. No.		Max Marks	CO
Q.1 A)	Illustrate the major phases in the Waterfall model of Software Development? Give any two advantages and disadvantages of Waterfall model? Which phase consumes the maximum effort for developing a typical software product? OR Discuss Scrum process model and Scrum framework.	10	CO1
Q.1 B)	Prepare Software requirement specification for Airline Reservation System.	10	CO1
Q.2 A)	What are the different types of risks that a typical software project might suffer from? Give examples of each type of risk? What are the strategies for risk mitigation?	10	CO2
Q.2 B)	For Safe home Information System estimate the effort and cost required using Function Point (FP). Suppose the organizational average productivity for system of this type =1.2 FP/PM and the cost per function point is Rs. 20,000. Assume all of the data are of simple complexity. 	10	CO3

Q.3 A)	You are required to design an enterance examination system. In this system, an applicant has to make the application by entering all the details and paying the fees for the aplication. The mode of payment is through credit/debit card or online. On the other hand, verification committee is responsible to verify each of the application. The system has an enrollment office which takes care of generating and dispatching the admit cards. On receiving the admit card, an applicant is typically allowed to sit for the examination. The examiner checks the answer sheets and submits the marks to the system. Based on the marks given by the examiner, the enrollment office produces the grade card and dispatches those to the applicants. Write all notations along with their semantics which are used in a Use case diagram and based on enterance examination system's information draw use case diagram and write usecase description of any two use cases.	10	CO3
Q.3 B)	Write all notations along with their semantics which are used in a Class diagram and Develop UML Class diagram for software that we need to develop to manage a library. Each library member has a name, date of birth, address, phone number and email. Each journal has a name, ISBN code, research areas, shelf in which located, and an editor. Each editor has a name and a nationality and publishes many journals. A journal may cover issues in several research areas. It is important to store the order of appearance of research areas, since these indicate their relative importance for the journal. It is important to keep track of the date on which a journal has been borrowed and returned, and the member carrying out the operation OR Write all notations along with their semantics which are used in a state machine diagram and Develop UML state machine diagram for arbitrarily deep nesting of parentheses?	10	CO3
Q.4 A)	Perform structured analysis (DFD level 0,1, and level 2) and structured design (structure Chart) for developing the Restaurant Automation System.	10	CO3
Q.4 B)	What is Design Pattern? Explain any two Generalised Responsibility Assignment Software patterns (GRASP).	10	CO3
Q.5 A)	What are the four elements that exist when an effective Software Configuration Management system is implemented? Discuss each briefly.	10	CO4
Q.5 B)	What do you understand by performance testing? Explain the different types of performance testing	10	CO5

****